Worksheet 14: L'Hôpital's Rule

MATH 1271

10/27/2026

1. For each limit below, answer the following questions:

   - Is the limit indeterminate?
   - If the limit is indeterminate, is the limit of the form where you can immediately apply L'Hôpital's Rule?
   - What is the value of the limit?

   (a) $\lim_{x \to 0} \dfrac{\ln(1+x)}{x}$

   (b) $\lim_{x \to \pi} \dfrac{\cos(x)}{x}$

   (c) $\lim_{x \to 0} \dfrac{\sin(x) - x}{\cos(2x) - 1}$

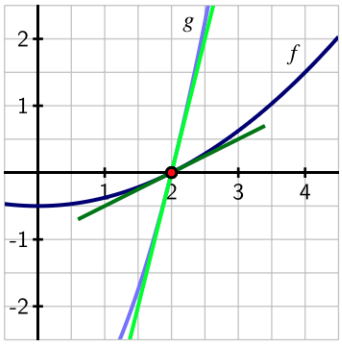
   (d) $\lim_{x \to 0^+} \dfrac{\ln(x)}{1/x}$

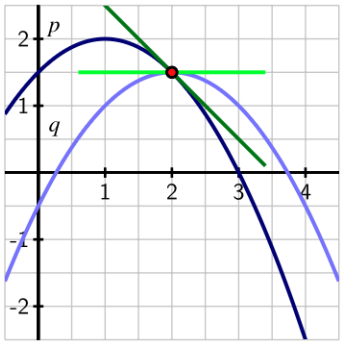
(e) $\lim_{x \to \pi/2} \dfrac{\tan(x)}{x - \frac{\pi}{2}}$

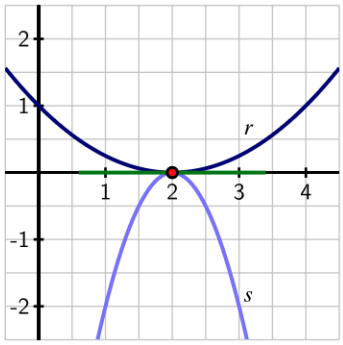
(f) $\lim_{x \to +\infty} xe^{-10x}$

(g) $\lim_{x \to 1} \left(\dfrac{1}{x-1} - \dfrac{x}{\ln(x)} \right)$

2. Three pairs of functions are graphed below:







(a) Use the left-hand graph to determine the values of $f(2)$, $f'(2)$, $g(2)$, and $g'(2)$. Then, evaluate $\lim_{x\to 2} \dfrac{f(x)}{g(x)}$.

(b) Use the middle graph to find $p(2)$, $p'(2)$, $q(2)$, and $q'(2)$. Then, evaluate $\lim_{x\to 2} \dfrac{p(x)}{q(x)}$.

(c) Assume that r and s are functions for which $r''(2) \neq 0$ and $s''(2) \neq 0$. Use the right-hand graph to compute $r(2)$, $r'(2)$, $s(2)$, and $s'(2)$. Explain why you cannot determine the exact value of $\lim_{x\to 2} \dfrac{r(x)}{s(x)}$ without further information being provided, but that you can determine the sign of $\lim_{x\to 2} \dfrac{r(x)}{s(x)}$. State what the sign of the limit will be (with justification).

3. **The Ultraviolet Catastrophe**: Before the advent of quantum mechanics, physicists studied the amount of energy given off by a radiating mass as a function of the wavelength of light emitted by it. We can use L'Hôpital's rule to get a hands on way of seeing how quantum mechanics remedies a terrible error!

   (a) **The Catastrophe**: The best pre-quantum mechanics description we had for this relationship between energy and wavelength was the Rayleigh-Jeans Law:

   $$B_{RJ}(\lambda) = \frac{K}{\lambda^4} \tag{1}$$

   where λ is the wavelength of light, and K is the *Boltzmann constant*. We call light that has a very small wavelength ultraviolet. Analyze the limit of the above function as $\lambda \to 0$ and use what you find to argue whether or not that function provides an accurate model for real world ultraviolet behavior.

   (b) **The Quantum Solution:** Maxwell Planck thought that assuming blackbodies could only emit highly specific energy levels of light (so called *quantized* energy states) would remedy this situation, and used that assumption to instead model blackbody radiation as:

   $$B_P(\lambda) = \frac{k_1}{\lambda^5} \frac{1}{e^{\frac{k_2}{\lambda}} - 1} \tag{2}$$

   where again λ is the wavelength and k_1, k_2 are just some constants. Analyze the limit of this function as $\lambda \to 0$ and use what you find to argue whether or not this function provides a more accurate model of real world ultraviolet behavior than the Rayleigh-Jeans model.